Progress Report 5

Topic: Visual Drilling Operations & Predictive Maintenance System

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Course: CSIS 4495 002 Applied Research Project

Section: 002

Team Lead: Uchechi Sebastian (Individual Project)

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Work Date / Hours Log

Date	Hours	Description
17-Jan-2026	3.5	- I explored and researched possible project ideas for the course, focusing on areas related to drilling operations and data analytics. I evaluated whether the project idea was feasible to implement within the course timeline and researched whether similar work had been done before. This helped me narrow down and decide on a suitable project that aligns with my background and course requirements.
20-Jan-2026	3	- I started learning React by reviewing the basics of components, JSX, and how a React application is structured. I focused on understanding how React will be used to build the frontend of my applied project.
21-Jan-2026	3.5	- I worked on Introduction of the project: Improving the introduction, stating the problem statement clearly, outlining what questions needs to be answered. - I searched for literature and research work based on my project to find similar research on my project. I searched for gaps within this research project work.
22-Jan-2026	2.5	- I continued literature research related to drilling operations, nonproductive time, and decision support systems. I reviewed how similar systems visualize operational data and noted limitations in existing approaches, particularly depth-based visualization. - I worked on my reference and included it in my project report after the research I did.
23-Jan-2026	3	- I worked on defining the project methodology, including data sources, data collection approach, and analysis techniques. - I also clarified how simulated drilling data would be used and documented the justification for using non-proprietary datasets in the project.
25-Jan-2026	1	- I created the project timeline for the complete project for the entire semester.

26-Jan-2026	3.5	- I drew the Gantt chart for the project timeline and included it in my proposal report. - I prepared the appendix. - I went through the entire report and did a lot of corrections and restructuring to make sure the proposal fits what my project is all about.
27-Jan-2026	2	- I set up the project GitHub repository structure according to the course requirements and organized folders for implementation and documentation. - I created the frontend, backend, and data directories under the implementation folder to support a clean development workflow.
29-Jan-2026	5	- I initialized the React frontend using Vite within the repository and verified that the development server runs correctly on my local machine. This step confirmed that the frontend environment was properly configured and ready for further development. - I continued learning React by implementing the initial application layout and navigation structure. I worked on creating reusable components for the application frame (layout and navigation) and began organizing the project into pages and components to follow best practices. I separated the HTML structure (JSX) and styling (CSS) into different files to improve readability and maintainability. I also installed and configured React Router to support page navigation between the wells list, dashboard, and reports sections of the application. - I implemented the well selection page and routing logic to allow navigation between different wells and their corresponding dashboard views. I tested navigation behavior to ensure that selecting a well correctly loads the appropriate dashboard page. During this process, I encountered and resolved common React

		development issues, including file naming inconsistencies, incorrect import paths, and component casing errors. I fixed these issues by standardizing file names, correcting relative import paths, and restarting the development server as needed, ensuring the application builds and runs successfully after all fixes.
02-Feb-2026	3.5	- Built and refined Wells page with color-coded well status cards - Implemented depth-based wellbore visualization - Added clickable depth segments with event details modal - Integrated routing between wells and dashboards - Improved frontend layout and interaction flow
05-Feb-2026	4	- I set up flexible column normalization so the upload can handle different Excel header names - I added validation for wells, operations, and events during the upload process - I added row-level error reporting to make debugging easier and catch bad data - I improved the duplicate checks so existing records aren't inserted - I added logging across the upload flow for better visibility - I finished wiring up the /upload endpoint with multipart file support - I updated the utils for mapping, normalization, and safer data parsing - I prepared the backend for future analytics by making sure uploaded data is clean and consistent

06-Feb-2026	2	- I created simulated datasets and collected sample datasets from different companies so I could compare how each one structures its drilling data. - I analyzed the differences in column names, formats, units, timestamps, and category labels across these datasets.
09-Feb-2026	5	- Implemented a complete rewrite of the Excel upload pipeline for wells, operations, and events. - Fixed all column-name mismatches between the transformer output and SQLAlchemy models. - Updated the Well insertion logic to use well_name instead of the invalid name argument. - Corrected the operations insertion to use operation_type instead of the non-existent op_type. - Updated event insertion and duplicate-check logic to use the correct model field (event_time). - I rebuilt the transform_dataset() function to ensure consistent normalization, depth conversion, timestamp parsing, and category mapping. - Ensured all transformed sheets contain the required columns before insertion. - Improved error reporting for wells, operations, and events to make debugging easier. - Cleaned up duplicate imports and removed unused or inconsistent logic. - Stabilized the entire ETL flow so uploads now run end-to-end with predictable behavior. -

13-Feb-2026	5	- Created operations router - Added GET /wells/{well_id}/operations endpoint - Added GET /wells/{well_id}/segments endpoint - Converted database operations into frontend-ready wellbore segments - Implemented basic risk-level classification logic - Connected router to main.py - Implemented DailyReport model and ingestion workflow for PDF uploads - Added duplicate detection using file hash and (well_id + report_date) - Created NNPC_FORMAT_A parser using pdfplumber with table-based extraction - Extracted operation rows from Operation Summary section - Implemented depth extraction logic (MD from / MD to detection) - Linked parsed operations to DailyReport and Well - Added debugging preview fields (matched_rows_preview, debug_preview). - Backend now supports: - Creating wells - Uploading daily drilling PDF reports - Preventing duplicate uploads - Parsing operation summary rows into database - Retrieving operations via GET endpoint
14-Feb-2026	4	- Added CORS + created /wells/{well_id}/dashboard endpoint and updated React Dashboards.jsx to fetch real segments/KPIs instead of hardcoded data. - Cleaned upload router to only handle PDF ingestion and moved wells/operations/segments endpoints under /wells routes to avoid duplicate Swagger entries. - Created wells router with create and list endpoints. Cleaned upload router to handle only report uploads.

		- Added CORS + created /wells/{well_id}/dashboard endpoint and updated React Dashboards.jsx to fetch real segments/KPIs instead of hardcoded data. - Fetch wells list from backend /wells/ Clicking a well navigates to /wells/:wellId dashboard
15-Feb-2026	4.5	- Added /reports router with list and detail endpoints - Enabled fetching all uploaded reports from DB - Enabled viewing a single report with its parsed operations - Registered router in main.py - Backend now supports a full Reports Page UI - Re-enable KPI light tone styling via tone classes Convert wellbore into fixed-height scroll view Add depth cursor slider + merge contiguous normal segments for cleaner visualization. - Implemented Reports.jsx to fetch /reports from backend - Added ReportRow.jsx component for list UI - Added Reports.css styling - Enabled navigation to individual report pages - Frontend now displays all uploaded drilling reports dynamically
16-Feb-2026	4	- Implemented a dedicated UploadReport.jsx page with styled form for uploading PDF reports. - Added UploadReport route in App.jsx and integrated upload button at the well level. - Added modal-based Create Well form directly inside Wells.jsx for quick well creation. - Redesigned Report Detail page to use a professional table view instead of cards. - Added new CSS: UploadReport.css and ReportDetail.css for a clean UI experience. - I Connected all frontend pages with backend routes properly. - Prepared UI structure for full multi-well report categorization.
19-Feb-2026	6	- Updated upload_daily-report FastAPI endpoint to use Form(...)

		for well_id, report_date, and parser_type. This ensures the frontend can upload reports using multipart/form-data. - Fixed 422 validation errors and restored successful PDF ingestion. - Added DELETE /reports/{report_id} endpoint in FastAPI - Added delete button to each report card in Reports.jsx - Added confirmation dialog and optimistic UI update after deletion - Added new CSS styles for delete button and improved card layout - Ensured seamless integration with existing upload and report detail flows. - fix(reports): register DELETE endpoint correctly - Moved delete_report() outside get_report_details() - Ensures FastAPI registers DELETE /reports/{report_id} - Resolves 405 Method Not Allowed during report deletion - main(wells): add Recently Viewed wells + Quick Actions bar + UI improvements - Added Quick Actions bar (Create Well, View Reports, Upload Report) - Added stats bar and modernized layout - Added color-coded well cards, operator avatars, and status badges - Improved overall dashboard experience - fix(wells): correct quick-action routing and add well-selection modal - View Reports and Upload Report now open a modal to choose a well - Added divider line to separate quick actions from wells - Improved navigation logic and UX consistency - style(wells): redesign Create Well modal for improved UX

		- Added modern modal card with animations - Improved input styling and labels - Added structured footer with Create + Cancel buttons - Enhanced spacing, shadows, and visual hierarchy
20-Feb-2026	4	- implement Sign Up page and full localStorage-based authentication - Added SignUp.jsx and SignUp.css - Updated Login.jsx to validate against stored user accounts - Added /signup route and integrated with login flow - Enabled full sign-up → login → dashboard experienc - added Sign Up link to Login page and complete auth flow - Added "Don't have an account? Create one" link - Linked Login → SignUp route - Updated styling for login switch link - Ensured smooth SignUp → Login → Wells navigation - Moved login/signup routes outside Layout wrapper - Added premium gradient background for auth pages - Optional glassmorphism card styling for modern UI
21-Feb-2026	2.5	- Replaced old icon with a cleaner well-bore-style SVG - Added new Home.css with modern gradient background and centered content - Updated Home.jsx to use new styling and improved structure - Ensured Home page loads without sidebar layout - Refined CTA button and feature pills for a more polished landing experience

23-Feb-2026	2	- Updated global App.css to ensure html, body, and #root all have width: 100% and height: 100% Prevented layout shrink issues caused by flex containers higher in the component tree Ensured that routed pages (e.g., Wells, Reports) now correctly expand to full available width Improved overall stability of the layout across all pages rendered inside <Layout/>
27-Feb-2026	4	- Fixed backend import error by correcting module paths in main.py(removed incorrect 'Implementation.' prefix) - Added predictive-maintenance API route with proper FastAPI router wiring - Introduced services/risk_engine.py for computing equipment risk scores - Ensured predictive.py imports use project-relative paths - Ensured predictive-maintenance results load dynamically in Maintenance.jsx - Cleaned router registration and removed duplicate predictive router include
4 -Mar-2026	6	**Dashboard & KPIs** - Fixed Event Count not updating by implementing proper cascade deletes. - Made all KPI cards clickable with short explanations. - Improved NPT KPI modal with a bar chart, proper labels, and correct report dates. **Report Detail** - Added editable mud and equipment tables under Operations, including mud descriptions. - Ensured Add/Edit buttons always show. - Resolved CORS/Vite proxy issues and added proper loading/error/editing states. **Backend** - Added mud_desc with automatic migration.

		- Added endpoints for updating equipment and adding mud. - Updated ingestion to store mud/equipment and return them in report details. **Parser (NNPC Format A)** - Extracted mud and equipment tables reliably using section-aware logic. - Prioritized tables with real data and expanded header keyword detection. **Wellbore Visualization** - Replaced fixed pipe height with dynamic height based on depth and segment count. - Removed CSS height constraints and prevented clipping by allowing overflow. **Equipment in Segment Modal** - Displayed equipment for each segment using equipmentByReport. - Passed equipment into SegmentModal and rendered a clean table when available. **AI Segment Analysis** - Added structured analysis output (reasons, factors, history, prevention, methodology). - Added endpoint for segment analysis and integrated it into SegmentModal with a full explanation view.
8 -Mar-2026	6	- Added optional LLM support for segment explanations with automatic fallback to rule-based logic. - Implemented prompt building and JSON-mode parsing for LLM responses. - Shortened and refined title-source text for clearer explanations. - Made both LLM and rule-based outputs more segment-specific using depth, operation, and NPT.

		- Removed generic contributing factors and simplified methodology text. - Connected the Maintenance page to the backend and removed all hardcoded data. - Added loading/error states and dynamic rendering for maintenance results. - Aggregated equipment by type on the backend so each appears only once. - Updated maintenance notes to show “Accumulated across N reports” when applicable. - Synced LLM and rule-based maintenance logic to reflect aggregated equipment data. - - Backend: Add GET /wells/summary endpoint returning per-well KPIs (depthMax, - nptHours, eventCount, criticalEvents, highRiskZones, maintenanceRisk) and - report_count for the Summary Reports page. - **Frontend:** - Added Summary Reports page at /summary-reports that lists all wells, with KPI cards, pie charts for NPT by well and maintenance risk distribution, and a "Download report (PDF)" button using the browser print dialog. - Added SummaryReports route in App.jsx and SummaryReports.css for layout and print-friendly styles. - Harden summary fetch with a 15s timeout and effect cleanup to avoid hanging when the backend is unavailable. - Fixed max() generator syntax in wells summary (parenthesize generator when using default=).

11 -Mar-2026	5	- Added "Well Location" nav link in Layout.jsx under Summary Reports (route /well-locations). - New WellLocations page: fetches wells from API, geocodes location/name via Nominatim, - displays OpenStreetMap (Leaflet) map with markers and popups linking to well dashboards. - Sidebar lists wells on map and wells not geocoded; map fits bounds to markers when present. - Used Leaflet only (no react-leaflet) to avoid React 19 peer dependency conflict. - Added WellLocations.css for map layout and responsive sidebar. - Added CSS variables in App.css for primary blue, sidebar, accents (success/warning/danger), and surfaces. - Added Inter font (Google Fonts); use for body and headings. - Redesigned sidebar: dark blue background (#0f172a), light text, active link with primary blue accent. - Used theme variables across Wells, Dashboards, Summary Reports, Home, Login, SignUp, Well Locations, Reports, Upload, KpiCard, and modals. - Aligned primary blue to #2563eb and use success/warning/danger accents for badges and risk states.
13-Mar-2026	3	- Improved rule-based segment analysis in ai_engine.py so normal segments and NPT events get distinct, data-driven titles that include operation type and depth range instead of all defaulting to "Segment Analysis". - Added a new POST /wells/segment-text-analysis endpoint that takes free-text description (e.g. from SegmentModal's Description section) and returns the same structured explanation shape used in the segment modal (titleSource, flaggedReason, contributingFactors, similarEventsInHistory, technicalFactors, preventionMeasures, methodology) by

		reusing analyze_segment. - Createded a backend .env placeholder for OPENAI_API_KEY so LLM-based analysis can be enabled without hardcoding secrets.
15 -Mar-2026	5.5	- I extended the backend POST /wells/segment-text-analysis endpoint to accept real segment attributes (depth_from, depth_to, operation_type, npt_hours, level), ensuring the analysis engine uses accurate segment data instead of default placeholders. - On the frontend, I updated the SegmentModal so it now sends the correct segment metadata (depth range, operation type/event type, NPT hours, and level), allowing the explanation header to display the true operational context. - I aligned the wellbore visualization with backend logic by using the API-provided segment.level for color coding, ensuring the red/warning states match the AI or rule-based classification rather than relying on frontend keyword heuristics. - I improved the segment analysis output by making the “Why Was This Flagged?” and “Why It Was Flagged Critical” sections fully level-aware and removed repeated descriptions to keep the explanations concise and segment-specific. - I added _get_llm_flag_explanation(segment) so the LLM can generate a short, natural explanation of what happened and why the segment was flagged (critical or warning), and integrated it into the flaggedReason and critical-flag cards when an API key is available. - For rule-based fallback, I refined _explicit_reasons_for_level() to produce clearer flagged text and removed unnecessary quoted descriptions from Stuck Pipe and Reaming contributing factors. - I updated the main LLM prompt to require a plain-language, segment-specific flaggedReason that summarizes what happened

		and why it was flagged, ensuring the explanation is accurate, readable, and grounded in the segment’s description.
16 -Mar-2026	4.3	- I added Recharts and lucide-react to the frontend to support richer analytics visualizations and consistent iconography across the dashboard. - I implemented a full NPT Analytics modal that opens when the Non-Productive Time KPI is clicked, including a structured header, a Download NPT Report button, four KPI cards, and a Recharts-powered pie chart comparing NPT vs Productive Time with a legend and summary table. - I expanded the analytics section with a bar chart for NPT by event type, line charts for daily and cumulative NPT trends, and a detailed NPT Events Breakdown table showing event type, hours, occurrences, percentage contribution, and averages. - I updated the dashboard so it now passes wellName, segments, and kpis into the KpiModal, and added full styling in KpiModal.css for the NPT header, KPI rows, pie chart legend, and events table, with a fallback to the original simple modal when NPT data is missing. - I extended the Event Count KPI to open a dedicated Event Analytics view, which mirrors the NPT layout and includes severity-based pie charts, event count by type, daily and cumulative trends, events by depth range, and a detailed event table; the Download Event Report option uses window.print for PDF export. - I refined the Detailed Event Breakdown so it now shows only critical and warning events, and updated the Event Count by Type bar chart to use severity-based colors (red, yellow, green) for clearer visual distinction.

19 -Mar-2026	5.7	- Added `frontend/src/pages/FleetWideReports.jsx` and `frontend/src/styles/FleetWideReports.css` to render the new Fleet-Wide Reports UI.
		- Fleet page fetches `/api/wells/summary`, then loads each active well's `/api/wells/{wellId}/dashboard` to compute fleet-wide charts (NPT comparison, cumulative trend, event type distribution, events per well, radar for top wells, and a detailed well summary table).
		    o Added export controls to the fleet report header:
		    o CSV download (table data)
		    o PDF export via `window.print()` (like the Summary Reports workflow)
		    o Full report export (JSON) of the computed fleet metrics.
		- Updated `frontend/src/pages/SummaryReports.jsx` to render the new fleet-wide report component (`FleetWideReports`) instead of the older single-page summary layout.

21 -Mar-2026	4.5	- Added frontend/src/utils/segmentEventType.js with getSegmentEventTypeLabel() to classify events from whyItMatters, operation/event type, and nptHours (stuck pipe, lost circulation, well control, flowline/surface lines, directional/flow issues, reaming, casing/trip/cement, equipment check, minor delay, NPT event, then title cased operation fallback). - SegmentModal: Event Type card uses getSegmentEventTypeLabel(segment) instead of segment.eventType; Operation Type still shows raw report value. Event Count modal (KpiModal) - Event Severity Distribution: Normal slice green (#16a34a); custom outside - labels (no clipped "Normal"); label colors match slices; 3-row legend with - correct swatches; summary rows for Critical / Warning /

		Normal. - Events by Depth Range: Normal bars green; Legend formatter so labels match series colors. - Detailed Event Breakdown: pill badges for Type (Critical/Warning/Normal) and Severity (High/Medium/Low); bold duration; zebra striping on tbody. - Added utils/severityDisplay.js and styles/SeverityBadge.css for reusable - severity pills. Fleet-Wide / Summary Reports (FleetWideReports) - New "Detailed Event Breakdown (All Wells)" table (critical/warning segments) with Well ID and same badge styling as the modal. - Detailed Well Summary Status column uses the same pill styling (red/amber/green). - Table zebra rows; centered badge columns; bold duration column. Note: backend/parser/db changes (if any) are separate from the above UI work.
23 -Mar-2026	4.1	**Backend** - Added owner_email to wells and created a SQLite migration to update existing databases. - Implemented auth_scope with X-User-Email requirement, including get_well_for_user and assert_report_owned for ownership validation. - Added optional BOOTSTRAP_WELL_OWNER_EMAIL to assign ownership to previously unowned wells during development. - Updated wells list/summary to filter by owner; well creation now sets owner automatically. - Added ability to claim an unowned well on POST with the same well_id.

		- Enforced ownership checks across dashboards, maintenance, reports, and segment-analysis routes. - Added ownership verification before ingesting uploaded data. - Updated reports and report-details to ensure listing, filtering, updating, and deleting all respect well ownership. - Applied the same ownership checks across all well-scoped routes in the operations router. **Frontend** - Introduced a centralized apiFetch() that automatically sends X-User-Email from localStorage. - Added 401 handling that clears the session and redirects to login when needed. - Replaced all direct /api fetch calls across Wells, Dashboard, fleet/summary reports, maps, maintenance, reports, report detail/edit, upload, and segment text analysis with the new authenticated fetch layer. - Updated logout flow to clear recentWells to prevent stale data after session changes.

Summary Description of work done during this reporting period

Last week, I expanded the reporting capabilities by building a full Fleet-Wide Reports interface, including new pages, styles, and export options. The fleet view now aggregates /api/wells/summary and each well's dashboard data to generate cross-well analytics such as NPT comparisons, cumulative trends, event-type distributions, radar charts, and a detailed well summary table. I also added CSV, PDF, and JSON export controls, updated the Summary Reports page to use the new component, and introduced utilities for classifying segment event types. Several UI components were upgraded with clearer severity and event-type displays, improved legends, consistent badge styling, zebra-striped tables, and reusable severity pill components.

On the backend, I implemented full well-ownership enforcement. This included adding owner_email to wells with a migration for existing databases, introducing an auth_scope that requires X-User-Email, and adding helper functions to validate ownership across all well-scoped routes. Wells are now filtered by owner, new wells automatically inherit the creator's email, and unowned wells can be claimed. Ownership checks now gate dashboards, maintenance, reports, segment analysis, uploads, and all operations routes. Report listing, filtering, updating, and deletion all respect the same ownership rules.

On the frontend, I centralized authentication by introducing an apiFetch() wrapper that automatically attaches X-User-Email and handles 401 responses by clearing the session and redirecting to login. I replaced all direct API calls across Wells, Dashboard, fleet and summary reports, maps, maintenance, uploads, report editing, and segment analysis to use this new authenticated layer. The logout flow was also updated to clear recentWells, ensuring no stale data persists between sessions.

Repo Check in of Implementation completed

I have checked my work to the GitHub repository according to the course requirements. The repository currently includes the following folders and files:

1. ReportsAndDocuments/
    - **Proposal/**
        - Project proposal document.
    - **ProgressReports/**
        - Progress Report #1
        - Progress Report #2
        - Progress Report #3
        - Progress Report #4
    - **MidtermReport/**
        - UchechiS_MidtermReport

2. Implementation/
    - **frontend/**
        - Initialized React frontend project using Vite
        - src/ directory containing:
            - pages/
                - Wells
                - Dashboard
                - Reports
                - Maintenance
                - SignUp.jsx
                - Login.jsx
                - Layout
                - SummaryReports.jsx
                - Maintenance.jsx
                - EditReport.jsx
                -
            - components/
                - KpiCard
                - KPIModal
                - ReportRow
                - SegmentModal
                - Wellbore
                - Layout
                - UploadBox
            - styles/ (separate CSS files for pages and components)
        - package.json and related configuration files
    - **backend/**
        - app/
            - __init__.py
            - main.py
        - models/

        - __init__.py
        - well.py
        - operation.py
        - event.py
      - routers/
        - __init__.py
        - wells.py
        - operations.py
        - events.py
        - analytics.py
        - risk.py
        - upload.py

      - schemas/
        - well_schema.py
        - operation_schema.py
        - event_schema.py
      - services/
        - injestion_service.py
        - ai_engine.py
        - kpi_services.py
        - risk_services.py

      - utils/
        - __init__.py
        - column_mapping.py
        - transform.py
        - database.py
    - **data/**
      - Folder created to store simulated or example datasets for later phases

3. Miscellaneous - Misc
4. README.md

AI Use Section

AI use Table

AI Tool	Version / Account	Specific Use	Value Added
ChatGPT	GPT-5.2	Assisted with refining the project idea, structuring the proposal, improving academic writing, and clarifying the research design, methodology, and timeline.	Helped organize thoughts clearly, improve academic clarity, and ensure the project scope and structure align with the course.
Copilot	Free	Summary of Relevant Literature and Existing Research	Helped to clearly identify the existing research on the project and showed how the project differs from various working projects.
ChatGPT	GPT-5.2	Used as learning support while developing the React frontend, including understanding React components, JSX structure, routing with React Router, file organization, and troubleshooting common setup and import issues during implementation.	Supported faster understanding of React concepts, helped resolve configuration and import errors, and improved confidence in implementing the frontend correctly within the project timeline.
Figma AI	Free	Used to generate UI layout suggestions and design inspiration for the drilling dashboard, including card layout, spacing, and visual hierarchy.	Helped to clearly identify the existing research on the project and showed how the project differs from various working projects.
Copilot (Backend Development)	Free	Assisted with setting up the FastAPI backend, configuring SQLite, creating database models, organizing folder structure, and planning the data-upload workflow.	Accelerated backend setup, ensured clean architecture, and improved clarity on how data flows from upload → database → analytics → frontend.

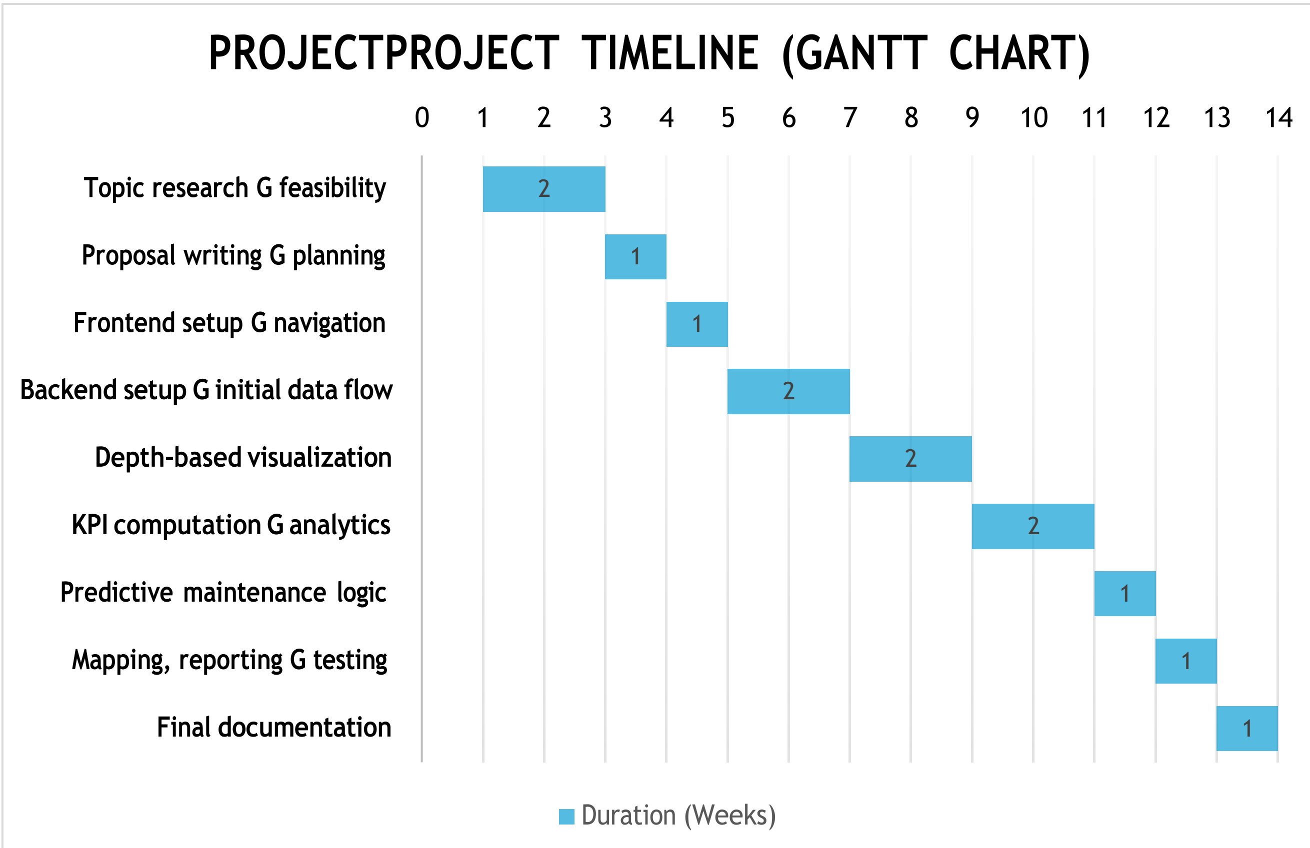
Copilot (Backend Development)	Free	I worked through the Excel upload pipeline, including column normalization, validation, duplicate handling, row-level error reporting, and debugging the upload endpoint.	It helped me break down the errors clearly, understand what was causing the failures in the upload pipeline, and refine the normalization and validation flow so the backend is more stable, predictable, and better prepared for the analytics features I'll build next.
Cursor	Chat	Implemented the Summary Repor ts feature: new backend endpoint GET /wells/summary (per-well KPIs and report count), new frontend page at /summary-reports with well cards, pie charts (NPT by well, maintenance risk distribution), and "Download report (PDF)" via print; added route and styles; fixed loading hang with a fetch timeout and backend syntax error (max() generator).	It helped me deliver full-stack feature (API, React page, styling, print layout) and fixed runtime/syntax issues so the Summary Reports page loads and the backend starts correctly.
Copilot	Free	Helped me design and implement the new Well Locations feature (Leaflet map, geocoding, responsive sidebar), refine the segment-analysis pipeline with real metadata + LLM/rule-based improvements, and build full NPT/Event analytics using Recharts, KPI cards, charts, and PDF export.	Enabled me to deliver multiple full-stack features faster, improve UI consistency, enhance analysis accuracy, and ship advanced analytics views that make operational insights clearer and easier to explore.
Copilot	Free	Helped me build the Fleet-Wide Reports system, cross-well analytics (NPT trends, event distributions, radar charts), improved segment classification, upgraded severity/UI components, and implement full well-ownership enforcement across backend routes	Enabled me to deliver complex reporting features faster, improve data accuracy and access control, streamline UI consistency, and ship a more polished analytics experience across the entire platform.

Project Planning and Timeline

Project Timeline

Phase	Week(s)	Milestones	Deliverables
Phase 1: Topic Exploration & Domain Research	Weeks 1 - 2	Explored potential project domains Reviewed drilling operations context Identified problem area and feasibility	Project idea selection Initial domain understanding
Phase 2: Proposal Development & Planning	Week 3	Defined project scope and objectives Identified data requirements Prepared a formal project proposal	Submitted project proposal Finalized research questions and methodology
Phase 3: Frontend Development (Core UI)	Week 4	Implement the main application layout Build well selection and navigation views	Initial React frontend Well selection interface
Phase 4: Backend Setup & Initial Data Flow	Weeks 5–6	Set up backend framework (FastAPI) Designed initial database schema Implemented basic API endpoints Connected frontend to backend using sample data	Functional backend service Initial API endpoints Frontend consuming backend data
Phase 5: Depth-Based Visualization	Weeks 7–8	Implemented vertical wellbore (pipe) visualization Mapped drilling events to depth intervals	Interactive depth-based visualization
Phase 6: KPI Computation & Analytics	Weeks 9–10	Compute NPT and event-based KPIs Aggregate data by depth and operation Enable downloadable drilling summary reports.	KPI analytics module Report on export feature
Phase 7: Predictive Maintenance Logic	Weeks 11	Implement rule-based risk indicators Link risk to recorded events and equipment	Predictive maintenance indicators with explanations
Phase 8: Mapping & Report Generation & Testing	Weeks 12	Implement a well location map view Test system functionality.	Location-based well view Tested application
Phase 9: Final documentation	Week 13	Prepare final documentation	Final project report

Gantt chat – Project timeline



Appendix

AI Prompt(s) Used

- Explain React components, pages, layout, and routing like I'm a beginner, and guide me step-by-step to set up a React app using Vite on Windows.
- I'm building a drilling dashboard UI. Help me create a Week 4 React layout with wells list + navigation. Keep CSS in separate files and explain what each file does.
- I'm getting an import/casing error in Vite/React. Diagnose the error and tell me exactly what files to rename and what import paths to use.
- How do I modify the frontend such that when I click on the view detailed explanation, it goes ahead to show me the detailed explanation and the analytics gotten based on the data provided? Please also explain very well so I can understand how it is done.
- No there is already <div className="footerBtns"> <button className="primaryBtn" type="button"> View Detailed Explanation </button> <button className="secondaryBtn" type="button" onClick={onClose}> Close </button> </div> so when you chick on the button that's where the three pictures come in.
- Okay, when I put the backend it will generate the analytics on its own without putting any explanations
- I want to put a button to show back to reports after uploading it is successful. And also, the depth is not working again it did not capture the dept correctly. Also, i would like us to put a full designed upload on the report side because the current upload is when you click on the wells and go to view report. but this time i want it directly on the report page under the wells page in the component part.
- so my wellbore pipe segment is meant to show different color codes when it detects that the report that was uploaded has any critical word in it, in the description but it is not showing .

Microsoft Copilot

- let us work based on my project. do I need the data to start so that I can do some analytics to connect it to the front end? Or let us start with, obviously you upload your dataset, then the app should receive this dataset and analyze it and show it on the wells page. When you then click on the particular well, it will show you the drilling pipe with different color coding. But first we should work on uploading the dataset first.
- For the uploading of files, these 3 categories, are you uploading the three documents as one csv, that is the; wells, operations, events. Also, most of this dataset have to be prepared after drilling right? that means there has to be some data input into their various tables. Or can the dataset from the company have everything, then the app begins to separate it by itself. That is to say, does it have to be prepared already.
- Good now, here in my wells, it looks so blank apparts from the wells that was created that is showing. What else do you think i can put here in the wells part to make it look more informative rather than showing only wells.
- when you click view report and upload report it shows page not found. can you link it to the exixting ones also move the quick bar to the side or draw a line to differenttiate it from the wells. Also i was thinking when you click on view reports you have to select the well first you want before seeing the reports. while upload report when you click on it, it can take us to the one we have already created.
- Okay now i see why the color codes are not showing on the segment. I tried editing one of the decription and i put pack-off inside so that it can display the color code. But i noticed the edit is not updated in the segment but it updated in the report details and backend. so the in the segment if you click the segment modal for each block it shows the original description but doesnt show the one i edited. so my testing did not work. Do you understand what i am trying to explain. so the segment is getting the description dirrectly from the uploaded report not minding if it was edited or not so it doesnt show my update. I edited it cause i just wanted to test the colors on the wellbore. Do you understand? and what can i do now?
- lets continue the upload i think most companies use excel not csv right so is it better to use excel to upload the data or csv based on what companies use.
- I dont want the exact names because not everyone in the industry would have time to put the exact names.
- okay if I integrate the script into the endpoint upload, that means it is ready for use. what does that mean. Also, after integrating it, can i test it to see if it is working. secondly will

the upload be able to carry thousands of roles of data.

- No i want it to show which report or date each NPT came from and put it in a bar chat. in the KPIModal for NPT. For example, the x axis will be the particular reportdate it occured, why the y axix will be the number of NPT. So when you click on the NPT KpiCard @Dashboards.jsx (95-108) , it goes to the modal and brings out the barchart of each report that has summed up to the total NPT like i have described.
- I think creating the edit button in the report details so i can edit the mud and Equipment table if it did not work correctly. And also be able to save it after editing. I also noticed the mud details did not have the first column - Mud desc. please include it and put the edit table for both.
- The pie chart labels for Normal are getting cut off how can I show the full label outside the slice?
- The legend colors don't match the donut for Warning vs Normal how do I align the legend with the slice colors?
- For the horizontal stacked bar 'Events by Depth Range', how do I change Normal from gray to green and make the legend text match the bar colors?
- How do I add pill-style badges for Type and Severity in the Detailed Event Breakdown table, with red/amber/green like a risk UI
- Right now every account sees the same wells after login how should I scope wells in the API so each user only sees their own?
- We only store the user email in localStorage what's a simple way to tell the backend which user is calling the API?

Cusor:

- I want to add a Summary Reports page that shows an overview of all wells on one screen so we can see what's going on per well and export it. I'd like to understand how to design this end-to-end: what backend data we need e.g. one summary endpoint vs many calls, how to present it e.g. cards per well, KPIs, and how to add simple charts e.g. pie charts for NPT or risk and a way to download the report for example, print to PDF. Can you walk me through the design and implementation so I can see how frontend and backend fit together?
- On the Summary Reports page, the UI stays on 'Loading summary…' and never finishes. I'd like to understand why that happens: is it usually because the request never completes for example, backend not running or proxy, and how should we handle

it in the frontend e.g. timeouts, error messages, and cleaning up when the user leaves the page so the app fails in a clear way instead of hanging?

- When I start the backend I get a SyntaxError on the line with max(o.depth_to or 0 for o in ops, default=0). I'm not sure why the generator expression is invalid here. Can you explain how max() works with a generator and a default argument in Python, and what the correct parentheses should be so I can fix it and avoid similar mistakes?
- In my drilling dashboard,the segment modal sometimes says 'no critical or warning key words were found'even when the segment is red.How can I make the Segment Analysis explanation title, 'Why Was This Flagged?use the same segment.level that drives the we llbore color so the text always matches the color?
- How can I have my backend AI analysis read the segment description and generate a short plain-language explanation of what happened and why it was flagged critical/warning, instead of just repeating the description or using a fixed template?
- If i previously implemented the NPT download as a CSV generator in KpiModal.jsx, what's the correct way to refactor it back to a window.print()-based PDF workflow so graphs are captured, and where exactly should I put that handler?
- How can i reuse my existing NPT Analytics modal CSS (header, KPI row, section cards, tables) for a new Event Analytics view without duplicating too much styling logic?
- How can i wire the 'Download Event Report' button in my Event Analytics modal so it uses window.print() (like my Summary Reports page) and includes all the event charts and tables in the generated PDF?
- What's the best way to pass wellName, segments, and kpis from my Dashboards.jsx page into a generic KpiModal so that clicking different KPI cards (e.g. NPT vs Event Count) can render completely different analytics views based on the same underlying dashboard data?